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18 July 1966
SUPERSEDING
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14 August 1964

MILITARY SPECIFICATION
CARTRIDGE, 7.62MM, NATO, BALL, DUPLEX, M198

1. SCOPE

1.1 This specification covers Cartridge, 7.62mm, NATO, Ball, Duplex, M198, intended for use in the M14 Rifle.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for porposal, form a part of this specification to the extent specified herein.

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-636	- Visual Inspection Standards for Small Arms Ammunition Through Caliber .50
MIL-STD-644	- Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing and Marking of Small Arms Ammunition
MIL-STD-1170	- Visual Standards and Comparison Methods for Evaluating Grain Configuration in 7.62mm Cartridge Cases

DRAWINGS

Munitions Command

D10534595	- Cartridge, 7.62mm, NATO, Ball, Duplex, M198
C7643674	- Classification of Cartridge Case Defects
F10534619	- Packing and Marking, Cartridges, 7.62mm, NATO, Ball, Duplex, M198; Cartons; Box, Ammunition, M2A1; Box, Wirebound

FSC 1305

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DRAWINGS (Cont'd)

F10535487 - Packing and Marking, Cartridges, 7.62mm, NATO, Ball, Duplex, M198, 5 Round Clips; Cartons; Bandoleers; Box, Ammunition, M2A1; Box, Wirebound
IEL-10534595 - Inspection Equipment List for Cartridge, 7.62mm, NATO, Ball, Duplex, M198

PUBLICATIONS

Munitions Command
AMCR 715-505, Volume 3 - Ammunition Ballistic Acceptance Test Methods, Volume 3, Test Procedures for 7.62mm Cartridges

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- The cartridge shall comply with the requirements specified on Drawing D10534595, referenced specification and the following:

3.1.1 The cartridge, after being subjected to a 1500 mile transportation simulation test, shall comply with the requirements of 3.6, 3.7, 3.8, 3.9, 3.10, 3.11, 3.12 and 3.13.

3.2 First article sample.- Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.3 Bullet extraction.- The force necessary to extract the front bullet from the cartridge case shall be not less than 40 pounds.

3.4 Residual stress.- The cartridge case shall not split when subjected to a one percent mercurous nitrate solution for 15 minutes.

3.5 Waterproof.- The cartridge shall not release more than one bubble of air when subjected to a pressure differential of 5 pounds per square inch (psi) for 15 seconds.

3.6. Accuracy and dispersion.- The average of the mean radii of all targets of the front bullets of the sample cartridges fired at 100 yards shall not exceed 2.0 inches. The average of the "circles of probable error" (as defined in Appendix A) of all targets of the same sample cartridges fired at 100 yards shall be not less than five (5) inches and not greater than ten (10) inches.

3.7 Action time.- The action time (overall primer ignition, propellant burning and front bullet-barrel time) of the cartridge shall not exceed 4 milliseconds.

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3.8 Velocity.- The average velocity of the front bullets of the sample cartridges, conditioned at 70° ± 2° Fahrenheit (F), shall be 2750 feet per second (ft/sec) plus or minus 30 ft/sec, at 78 feet from the muzzle of the weapon. The standard deviation of the velocities of the front bullets shall not exceed 32 ft/sec. The velocity of each rear bullet of the same sample cartridges shall be not less than 2,200 ft/sec at 78 feet from the muzzle of the weapon.

3.9 Chamber pressure.- The average chamber pressure of the sample cartridges, conditioned at 70° ± 2° F, shall not exceed 52,000 pounds per square inch (psi). Neither the chamber pressure of an individual sample cartridge nor the average chamber pressure of the sample cartridges plus three standard deviations of chamber pressure shall exceed 58,000 psi.

3.10 Port pressure.- The average port pressure of the sample cartridges conditioned at 70° ± 2° shall be not less than 8,500 psi.

3.11 Penetration.- The front bullets of the cartridge shall pass completely through a 10 gage (0.135 inch) mild steel plate (SAE-1010 or SAE-1020 with a hardness of Rockwell B55 to Rockwell B90) oriented at 90 degrees to the line of fire at a range of 300 yards.

3.12 Temperature stability.- When the sample cartridges are subjected to the following storage conditions, the average velocity of the front bullets at 78 feet from the muzzle of the weapon shall not increase by more than 150 ft/sec or decrease by more than 250 ft/sec with respect to the average velocity at 78 ft. from the muzzle of the weapon of the sample cartridges of the same lot conditioned at 70° ± 2° F, the velocity of each rear bullet shall be not less than 2,000 ft/sec at 78 feet from the muzzle of the weapon, and the average chamber pressure shall not increase by more than 5,000 psi or decrease by more than 15,000 psi with respect to the average chamber pressure of the sample cartridges of the same lot conditioned at 70° ± 2° F.

Condition
Stored at -65° ± 5° F for not less than 6 hours and fired at that temperature.
Stored at 125° ± 2° F for 4 hours and fired at that temperature.

3.13 Function and casualty.- The cartridge shall function without casualty at ambient temperature and under the conditions specified in paragraph 3.12.

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3.14 Grain configuration.- The grain configuration of the sidewall of the finished cartridge case shall fall within the range defined by the grain configuration standards illustrated in MIL-STD-1170, Figures 1 through 6.

3.15 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.15.1 Metal defects.- The cartridge shall be free of folds, wrinkles, deep draw scratches, scaly metal, dents and other defects. In addition, the copper plating on the bullet shall be smooth, fine grained, adherent, free from blisters, pits, nodules, or unplated areas.

3.15.2 Foreign matter.- The cartridge shall be free of corrosion, stains, discolorations, dirt, oil, and smears of lacquer.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 1500 cartridges. The sample shall be manufactued using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

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4.2.1.2 Initial production sample failure.- Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Presentation of lot.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- The ammunition lot size shall be as agreed upon between the supplier and the contracting officer, provided the formation of the lot is in accordance with MIL-STD-105.

4.3.1.3 Lot identification.- Each lot of ammunition shall be identified as to type, caliber and model, as well as with a lot number and the manufacturer's identification as assigned by the procuring activity. Each lot shall be further identified by a Federal Stock Number assigned by the procuring activity.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a class basis in accordance with the classification of defects, Table I, using applicable sampling plans and acceptance criteria of MIL-STD-105. The Acceptable Quality Level (AQL) for the major class shall be 0.25 percent and the AQL for the minor class shall be 1.50 percent.

4.3.2.1 Classification of defects.- The classification of defects shall be as specified in Table I.

Table I

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Visual 1/				
	Cartridge				
1	Discolored, dirty, oily, smeared			X	
2	Corroded or stained, if etched		X		
	Case				
4	Round head		X		
5	Dent				X
6	Split case				
	in K, L, or M location	X			
	in I, S, or J location		X		

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
7	Perforated case	X			
8	Draw scratch				X
9	Scratch			X	
10	Beveled underside of head		X		
11	Case mouth not crimped in cannelure		X		
12	Scaly metal				X
13	No chamfer on head (rim)		X		
14	Fold			X	
15	Wrinkle			X	
16	Buckle			X	
17	Bulge			X	
18	Illegible or missing head stamp			X	
19	Defective head			X	
20	Defective mouth			X	
21	No visible evidence of mouth anneal		X		
	Bullet				
22	Dent			X	
23	Scratch			X	
24	Split bullet		X		
25	Loose bullet		X		
26	Missing cannelure		X		
27	Scaly metal (bullet)				X
28	Upset (crooked) point			X	
29	Missing or defective plating		X		
30	Blunt point			X	
31	Defective cannelure			X	
	Primer				
32	No primer	X			
33	Cocked primer	X			
34	Inverted primer	X			
35	Loose primer		X		
36	Nicked or dented primer			X	
37	No waterproofing material (primer pocket joint)			X	
38	Defective crimp			X	

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Table I (Cont'd)

No.	Defect and Method of Inspection	Critical	Major	Minor	Major or Minor
	Gaging				
39	Total length		X		
40	Cartridge profile failure (requiring more than 20 lbs dead weight to insert in profile and alignment gage)		X		
41	Diameter of extractor groove, max.		X		
42	Diameter of extractor groove, min.			X	
43	Diameter of head		X		
44	Thickness of head		X		
45	Length to shoulder datum		X		
46	Depth of primer		X		
	Weighing				
47	Weight, min. 2/		X		

1/ Refer to MIL-STD-636 for visual defect standards for defects 1 through 38 with the exception of Defects 24 and 29.

2/ Each lightweight cartridge shall be disassembled and the propellant weighed. Each such cartridge found to contain less than 25 grains of propellant shall be classed as a critical defect.

4.3.3 Tests.- The tests listed in Table II shall be conducted in accordance with the methods and procedures specified in 4.4.

4.3.3.1 Test samples.- The quantities for the various tests shall be as specified in Table II. Only cartridges having met the visual, dimensional and weight requirements shall be used in the ballistic tests, and shall have been selected in such a manner that the sample is representativeof the entire lot. The cartridges shall be thoroughly mixed before being divided into samples for the various tests.

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Table II

Test	Number of Cartridges: Ambient Temp.	70° ± 2° F	-65° ± 5°	125° ± 2° F	Requirement Paragraph
Bullet extraction 1/	25				3.3
Residual stress (Mercurous nitrate 1/	50				3.4
Waterproof 2/	50				3.5
Accuracy and Dispersion 3/	90				3.1.1 & 3.6
Action time 1/		50			3.1.1 & 3.7
Velocity 4/		20	20	20	3.1.1, 3.8 & 3.12
Chamber pressure 4/		20	20	20	3.1.1, 3.9 & 3.12
Port pressure 3/		20			3.1.1 & 3.10
Penetration 5/	10				3.1.1 & 3.11
Function & casualty 6/					3.1.1, 3.12 & 3.13
Rifle, M14	240		240	240	
Grain Configuration 1/	10				3.14
Hardness 3/	10				Drawing

1/ Failure of two or more cartridges to comply with the applicable requirement shall be cause for rejection of the lot. If one cartridge fails in the first test, a second sample consisting of double the number of cartridges in the first sample may be tested. If any failing cartridges are found inthe second sample, the lot shall be rejected.

2/ Failure of ten or more cartridges to comply with the applicable requirements shall be cause for rejection of the lot. If more than three but less than ten cartridges fail in the first test, a second sample consisting of double the number of cartridges in the first sample shall be tested. The lot shall be rejected if, in the combined first and second sample, ten or more cartridges fail to comply with the applicable requirement.

3/ Failure of the cartridges to comply with the applicable requirements shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test. Failure of the cartridges in the second sample to comply with the applicable requirements shall be cause for rejection of the lot.

4/ Failure of the cartridges in any sample to comply with the applicable requirement shall be cause for rejection of the lot subject to testing of a second sample consisting of double the quantity of cartridges used in the first test for the temperature or temperatures at which the failure occurred. Failure of the cartridges of the second sample to comply with the applicable requirement shall be cause for rejection of the lot.

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5/ This test shall be conducted on the initial production sample only.

6/ The lot shall be rejected when the function and casualty defects, plus all other firing defects, exceed the acceptance number for the cumulative sample in Table III. If the number of defects found in the first test exceed the acceptance number for the first sample, but is equal to or less than the acceptance number for the cumulative sample, a second sample consisting of double the quantity specified under function and casualty test shall be fired in the service weapon specified therefor. If the total number of defects in the combined first and second samples exceeds the acceptance number for the cumulative sample, the lot shall be rejected. If in testing a second sample, defects other than those for which the second sample is being tested should occur to the extent that they exceed the acceptance for the cumulative sample, the lot shall be rejected.

4.3.3.2 Firing defects.- Firing defects and acceptance numbers shall be as specified in Table III.

Table III

Defects	Acceptance First Sample	Acceptance Cumulative (1st & 2nd Sample
1. Misfire	1	2
2. Bullet remaining in bore 1/	0	-
3. Primer leak		
a. Perforation in firing pin indent in primer cup 2/	0	-
b. Escape of gas through primer cup other than 3a	1	2
c. Escape of gas around primer cup 3/	28	56
d. Blown primer or primer falls out of pocket on retraction of bolt	0	1
e. Primer remains in pocket but is physically loose	1	2
4. Case casualties		
a. Longitudinal split 4/		
(1) Neck and shoulder (I or S)	19	39
(2) Body (J)	2	4
(3) Body (K)	1	2
(4) To head (L)	0	1
(5) Through head (M)	0	1
b. Circumferential rupture 4/		
(1) Partial, shoulder or body (J&S)	1	2
(2) Partial, body (K)	0	1
(3) Partial, head (L)	0	1
(4) Complete	0	1

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Table III (Cont'd)

Defects	Acceptance First Sample	Acceptance Cumulative (1st & 2nd Sample)
5. Failure to extract	0	1
6. Weapons stoppage 5/	0	1

1/ No second sample permitted. Lot shall be rejected.

2/ If one or more defects are found in the first sample, a second sample consisting of double the quantity of cartridges specified under Function & Casualty of Table II shall be fired. Prior to the testing of the second sample, the firing pin of the specific rifle(s) in which the defect originally occurred shall be replaced by a new firing pin. If an additional primer perforation is found in the second sample, the lot shall be rejected.

3/ Of the cartridges in which this defect occurs, not more than 9 in the first sample nor more than 17 in the cumulative sample shall exhibit escape of gas around 50% or more than 50% of periphery.

4/ For location of defects indicated by letters in parentheses, see Drawing C7643674.

5/ All stoppages attributable to the ammunition, with the exception of misfire, complete rupture or failure to extract, observed in all tests shall be included.

4.3.4 Packaging, packing and marking inspection.- During or immediately prior to the packaging operation, 100% examination of the cartridges shall be performed to ascertain that the cartridge type conforms to the drawing. Occurrence of a high pressure test, dummy, blank or grenade cartridge shall be classed as a critical defect. All non-conforming cartridges shall be rejected. Inspection for packaging, packing, and marking shall be in accordance with MIL-STD-644 as applicable to the drawing.

4.3.5 Inspection equipment.- The examination and tests shall be made using the equipment listed on IEL-10534595, except as specified in 4.3.

4.4 Test methods and procedures.

4.4.1 Bullet extraction.- The cartridges shall be tested in an approved bullet extractor machine. The rate of travel of the test head shall be not less than three nor more than six inches per minute. The test shall be conducted in accordance with AMCR 715-505, Volume 3.

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4.4.2 Residual stress (Mercurous nitrate).- The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.3 Waterproof.- The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.4 Transportation simulation.- Prior to performing the tests of paragraphs 4.4.5 through 4.4.12, the sample cartridges for those tests shall be packaged as specified in the contract and packed in metal boxes. The metal boxes containing the cartridges shall then be vibrated at a rate of approximately 370 cycles per minute (cpm) with an amplitude, peak to peak, of 9/16 ± 1/64 inches for 60 minutes. The metal boxes shall be clamped to the vibrating surface to insure that the entire vibratory motion is imparted to the contents of the metal boxes. For the first 30 minutes the metal boxes shall be placed bottom down to the vibrating surface and for the final 30 minutes, the metal boxes shall be placed either side down. The sample cartridges shall then be removed from the metal boxes, unpackaged, and thoroughly mixed. The samples for the respective tests shall be drawn from this mix at random.

4.4.5 Accuracy and dispersion.- The test shall be conducted in accordance with the accuracy test procedure of AMCR 715-505, Volume 3. The front or leading bullet shall be coated with a colored substance that will leave an identifying mark on the target. The evaluation of the results for dispersion shall be in accordance with Appendix A.

4.4.6 Action time.- The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.7 Velocity.- The test shall be conducted in accordance with AMCR 715-505, Vol. 3, and 4.4.11, except that the velocities of both the front and rear bullets shall be recorded. The velocities of the rear bullets shall not be corrected.

4.4.8 Chamber pressure.- The test shall be conducted in accordance with Appendix B and 4.4.11.

4.4.9 Port pressure.- The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.10 Penetration.- The test shall be conducted in accordance with AMCR 715-505, Volume 3.

4.4.11 Temperature.- The tests shall be conducted in accordance with AMCR 715-505, Volume 3. The weapon or weapons in which these tests are fired shall be at room temperature.

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4.4.11.1 Low temperature.- The test sample shall be stored at minus 65 degrees F, plus or minus 5 degrees, for not less than 6 hours and shall be fired at that temperature.

4.4.11.2 High temperature.- The test sample shall be stored at 125 degrees F, plus or minus 2 degrees, for 4 hours and shall be fired at that temperature.

4.4.12 Function and casualty.- The tests shall be conducted in accordance with Publication AMCR 715-505, Volume 3 and 4.4.11. Two rifles shall be used, and the following number of cartridges shall be fired in each:

Ambient temperature	120
Low temperature	120
High temperature	120

4.4.13 Grain configuration.- The test samples shall be prepared and evaluated in accordance with MIL-STD-1170.

4.4.14 Hardness testing.- The bullets shall be extracted, the propellant removed and the primers extracted. Each cartridge case of the sample shall be prepared and placed on the appropriate test fixture for testing in accordance with Federal Test Method Standard No. 151, Method 244.1. The average of the hardness values of the sample cases for each prescribed point shall be computed and charted in accordance with the drawing requirements.

4.4.15 Defect penalty.- In any ballistic test, except function and casualty, in which the occurrence of a firing defect prevents the obtaining of a reliable result for the characteristic being tested, an additional shot shall be fired. That particular test shall not be penalized, but the acceptance or initial production sample shall be penalized for such defects in accordance with Table III.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.- (Worldwide shipment) The cartridges shall be packed in accordance with Drawing F10534619 or F10535487.

5.2 Marking and labeling.- Packing boxes shall be marked and labeled in accordance with the applicable drawing cited in 5.1.

6. NOTES

6.1 Ordering data.- Invitations for bids and contracts or orders will specify the following:

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6.1.1 Title, number and date of this specification.

6.1.2 Type and level of packing.

6.1.3 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.1.4 Provision for the submission of acceptance inspection reports containing final inspection results for each lot of ammunition presented to the Government.

6.1.5 Provision for the identification and submission of grain configuration photomicrographs prepared in accordance with MIL-STD-1170 where such submission is required.

6.1.6 Requirement for contractor to provide and maintain an inspection system in accordance with MIL-I-45208, Inspection System Requirements.

6.2 Document availability.

6.2.1 Copies of MIL-STD-1170, required for evaluating grain configuration may be obtained on a need-to-have basis through the Commanding Officer, U. S. Army, Frankford Arsenal, Philadelphia, Pennsylvania 19137.

Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1305-A468

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APPENDIX A

Dispersion Evaluation Procedure (7.62mm, Ball, M198)

1.0 CENTER OF IMPACT (MEDIAN METHOD)

1.1 The Center of Impact of each target consisting of ten (10) rounds shall be determined as follows:

1.1.1 Locate the ten (10) shot-holes made by the leading or front bullets. Identification is made possible by the marker left on the target by the identifying compound applied to each front bullet immediately before firing (e.g lithographic ink).

1.1.2 Draw a vertical line through the ten (10) shot-holes, so that five (5) holes are on each side of the line. This line shall be drawn so that it is equidistant between the two nearest shot-holes of the two groups. Similarly, draw a horizontal line such that five (5) holes are on each side of the line.

1.1.3 The intersection of the vertical and horizontal lines is the Center of Impact.

2.0 DISPERSION

2.1 Circle of Probable Error (CEP)

2.1.1 The CEP is the radius of the smallest circle which contains all of the leading or front bullets and one-half of the trailing or rear bullets, the center of which shall be the Center of Impact, determined as prescribed in 1.0. The circle is drawn so that it just encloses the outer edge of the applicable trailing or rear bullet (See Figure 1).

2.1.2 The radius of the circle is then measured and reported as the value of CEP for the target.

2.2 Recording of Results.

2.2.1 The CEP for each target.

2.2.2 The average CEP of all targets.

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APPENDIX A (Cont'd)

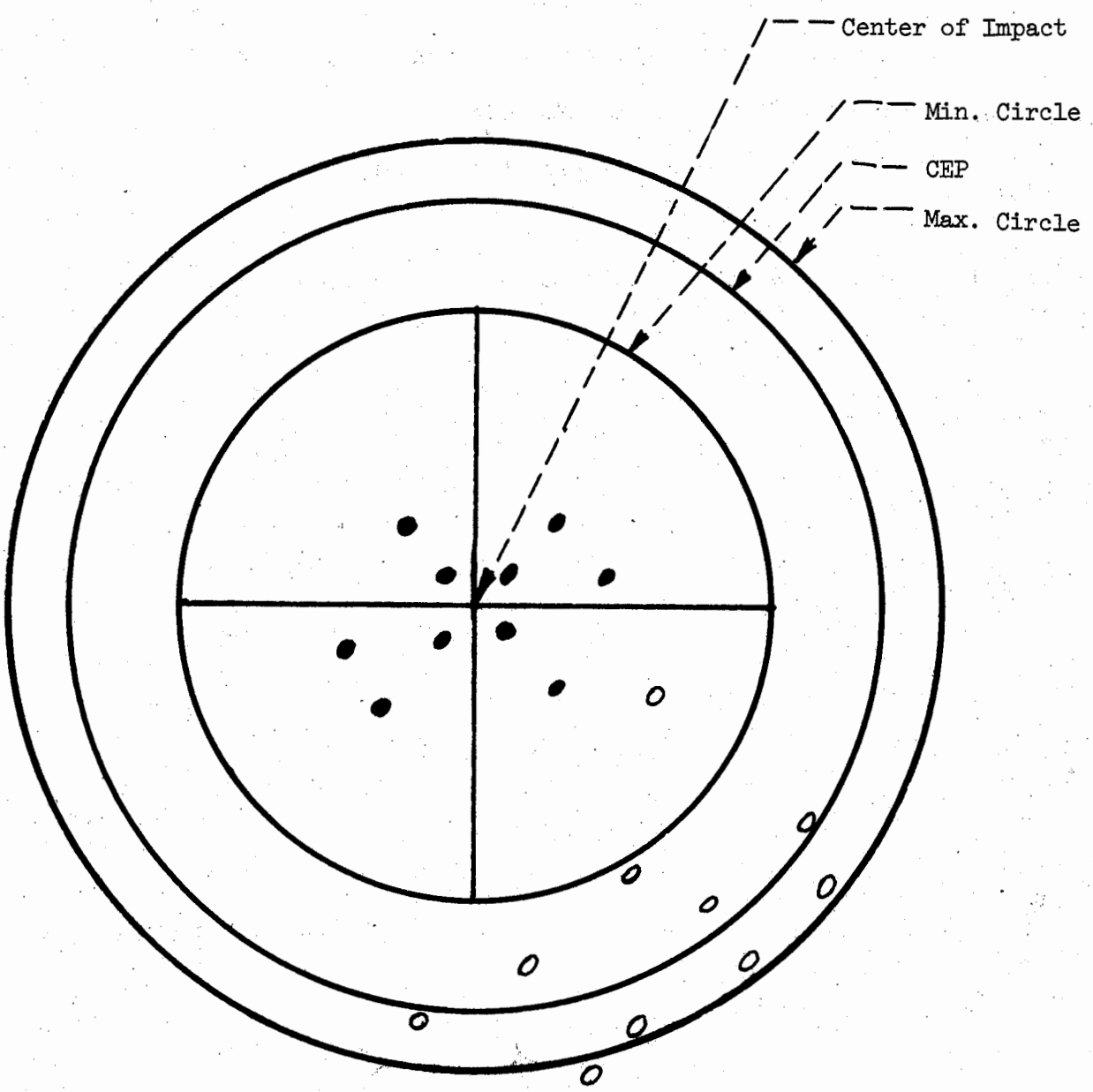


FIGURE 1

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APPENDIX B

Chamber Pressure Test Procedure (7.62mm, Ball, M198)

1.0 EQUIPMENT

1.1 Equipment listed in the Chamber Pressure section of the appropriate Inspection Equipment List shall be used.

1.2 The minimum distance between the disjunctors for recording velocity shall be fifty (50) feet.

1.3 Pier or mount on which the test fixture assembly is mounted shall be of solid construction.

2.0 USE OF REFERENCE CARTRIDGES

2.1 Reference cartridges shall be used to establish range and equipment corrections prior to firing any ammunition lot for acceptance. One reference cartridge shall be fired for each cartridge of test sample to be fired, up to twenty (20) cartridges. When the test consists of more than twenty (20) but not more than forty (40) cartridges, then twenty (20) reference cartridges shall be fired. When the test consists of more than forty (40) cartridges, one (1) reference cartridge shall be fired for every two (2) test cartridges.

2.2 After the required number of reference cartridges has been fired, the actual average chamber pressure of the reference cartridges shall be compared with the assessed value. If the assessed value is higher than the actual average chamber pressure of the refence cartridges, the difference is a plus correction and shall be added to the average chamber pressure of the test cartridges. If the assessed value is lower than the actual average chamber pressure of the reference cartridges, the difference is a minus correction and shall be subtracted from the average chamber pressure of the test cartridges. If the assessed value and the actual average chamber pressure of the reference cartridges are identical, then no correction is applied.

2.3 The pressure-barrel assembly shall be acceptable provided the results obtained with the reference cartridges are within $\pm$ 3500 PSI of the assessed value.

2.4 It is suggested that control charts be maintained, for record purposes, of the results from each barrel in which reference cartridges are fired.

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3.0 MEASUREMENT OF COPPER PRESSURE CYLINDERS

3.1 The chamber pressure shall be determined by using copper pressure cylinders. One cylinder shall be used for each cartridge to be fired.

3.1.1 The copper pressure cylinders shall be measured individually using a properly calibrated micrometer graduated to 0.0001 inch. The cylinders shall be measured prior tofiring, measurements recorded and the cylinders placed in recessed holding block in such a manner that their identity is maintained throughout the test.

3.1.2 Upon completion of the firing, the cylinders shall be measured again, and the decrease in length obtained by subtraction of the actual readings. The decrease in length obtained for each cylinder shall then be applied to the appropriate tarage table (a tarage table is supplied with each box of copper pressure cylinders) and the corresponding chamber pressure (PSI) shall be recorded on the test report form in such a manner that the velocity of each cartridge can be identified with the corresponding chamber-pressure obtained.

3.1.3 Whenever a new supply of copper pressure cylinders is received involving the use of different tarage tables, they shall be tested in comparison with the cylinders being replaced by firing forty (40) reference cartridges for chamber pressure in a pressure-barrel assembly of known characteristics, using twenty (20) of the new cylinders in alternation with twenty (20) of the old cylinders. The mean chamber pressure obtained with the new cylinders shall not vary by more than $\pm$ 1500 PSI from the mean obtained with the new cylinders shall not vary by more than $\pm$ 1500 PSI from the mean obtained with the old cylinders. If the means vary by more than $\pm$ 1500 PSI, the new lot shall be considered unacceptable and reported to the responsible agency.

4.0 TEST PROCEDURE

4.1 Pre-firing (Preparation for test)

4.1.1 The reference and test cartridges shall be placed in separate recessed holding blocks, primer-end down. Each sample, properly identified, shall be permitted to come to a temperature of 60° F. to 80° F. prior to being placed in the temperature-controlled container or room. The cartridges shall be placed in the temperature-controlled container or room in such a manner that all cartridges are subjected to a uniform temperature for a minimum of two hours, prior to firing. The container or room shall be maintained at a temperature of 70° F., $\pm$ 2° F., with a relative humidity of 60, $\pm$ 5 percent, and be of sufficient capacity to allow free circulation of air.

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4.1.2 The pressure barrel assembly is assembled in the test fixture on the mount. The chamber and bore of the barrel are wiped dry. The barrel is then boresighted into position.

4.1.3 The pressure barrel assembly shall be in accordance with the applicable drawings, if the "no-go" gages enter either the upper or lower end of the piston hole, the barrel assembly shall be disqualified.

4.1.4 The following measurements shall be made before a pressure barrel assembly is placed in service, after misfires, pierced primers, flowbacks, or whenever a part replacement has been made in the bolt assembly:

	Limits
Firing-pin protrusion	.060" - .075"
Firing-pin indent	.020" - .025"
Headspace (to 0.400" datum dia.)	1.628" - 1.631"

4.1.5 The velocity disjunctors are checked for position. It is of the utmost importance that the disjunctors be placed in their proper positions; measurements must be accurate.

4.2 During firing.

4.2.1 The long piston is placed in the piston hole but not fully seated until after the cartridge is in the chamber. Five (5) warmer (fouling) shots shall be fired. Velocity readings should be recorded by the chronographer to assure that the velocity-measuring equipment is functioning properly.

4.2.2 The recessed holding block containing the individually measured copper pressure cylinders, a sufficient quantity of obturating cups (a new obturating cup shall be used for each cartridge to be fired) and the short piston should be placed at a point convenient to the technician.

4.2.3 The recessed holding block containing the reference cartridges is removed from the controlled-temperature container or room and placed at a point convenient to the technician provided the temperature of the firing room is 70° F., ± 5° F., otherwise the cartridges shall be placed in an insulated box which has been conditioned at 70° F., ± 5° F., and the box placed at a point convenient to the technician; the cartridges are then removed singly from the insulated box immediately before firing. If an insulated box is not available, then the cartridges shall be removed singly from the controlled-temperature container or room immediately before firing.

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4.2.4 The dummy cartridge (obturator catcher) shall be placed in the chamber of the pressure barrel, making certain the opening in the dummy cartridge case is not aligned with the piston hole. The obturating cup shall then be placed in the piston hole, mouth-end down, and partially seated; using the short stem of the knockout tool. The shank of the pressure piston shall be dipped in oil and the oil then be allowed to drain from the piston. The drop of oil adhering to the bottom of the piston shank shall then be removed by touching the bottom of the piston shank to a cloth cleaning patch. The piston shall then be inserted into the piston hole and pressed down on the obturator until the piston has reached its correct final position. The head of the piston and the bottom of the anvil shall be wiped dry and free of oil. The copper cylinder shall be put in place and centered between the head of the piston and the bottom of the anvil. The anvil shall be screwed down lightly on the cylinder, using the thumb and forefinger. The dummy cartridge (obturator catcher) shall then be removed from the chamber.

4.2.5 In order that the propellant shall be uniformly positioned from shot to shot, attention to detail is necessary in handling and chambering the cartridge. The cartridge shall first be held vertically, bullet upward. It shall then be rotated slowly in a vertical plane, stopping the rotation momentarily after 180° of rotation when the bullet is downward, and then continuing through the remainer of 360° , stopping with the cartridge again bullet-end upward. The bullet-end of the cartridge should now be lowered slowly to a position slightly above horizontal and chambered very carefully, taking care that the primer-end of the case is not elevated above the bullet-end of the case. (The object is to have the cartridge seated in the chamber ready to fire, with the peopellant in a loose condition at the primer-end of the case, and with such airspace as is present, at the bullet-end of the case.)

4.2.6 The breech-block shall be closed gently. If the technician encounters any difficulty closing the breech-block or engaging the trip lever, the test shall be discontinued until such difficulty is corrected. If any delay occurs after the cartridge is place in the chamber, and the duration of the delay is such that the temperature of the cartridge has changed significantly, that cartridge shall be extracted and another inserted in its place.

4.2.7 The final tightening of the anvil on the copper cylinder shall now be accomplished. The proper torque to be applied to the thumb-screw is about one pound-inch. (This torque can be estimated with satisfactory accuracy by an experienced technician; to familiarize inexperienced technicians with the desired degree of tightness, a torque-measuring devide may be employed. This can be accomplished by drilling and tapping (threading) an axial hole in the knurled head of the anvil (thumb-screw), and inserting a bolt or screw to which a torque-measuring wrench can be attached. The usefulness of the anvil (thumbscrew) is not impaired by this modification.)

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4.2.8 The trip lever, to which the lanyard is attached, shall be engaged gently with the hammer. The technician should retire to a safe position and pull the lanyard with a smooth firm motion. The velocity of the shot shall be recorded by the chronographer.

4.2.9 The breech-block shall be opened, the fired case extracted and visually examined by the technician for possible case casualties. The copper cylinder shall be removed and placed in the recessed holding block. The piston shall be removed from the piston hole. The dummy cartridge (obturator catcher) shall be inserted into the chamber with the drilled hole aligned with the piston hole, the obturator and case fragment are then forced into it with the stem of the knockout tool. The dummy cartridge (obturator catcher) is then rotated in the chamber so that the drilled hole is no longer aligned with the piston hole. The procedure prescribed in 4.2.4 shall be then repeated. Upon removal of the dummy, the chamber and bore shall be checked for the possibility of any obstruction remaining in the barrel.

4.2.10 The procedure prescribed in 4.2.5 through 4.2.9 shall be repeated until the required number of reference cartridges has been fired.

4.2.11 Continuous air cooling should be used on the barrel. If air cooling is not available, barrel shall be allowed to cool to ambient temperature between each series of tests, or after a maximum number of twenty (20) cartridges has been fired. At no time shall the exposed metal surface of the test barrel become too hot to grasp with the bare hands (approx. 140° F.).

4.2.12 The copper pressure cylinders used with the reference cartridges shall then be measured and the measurements recorded in accordance with 3.1.2. The chamber pressure correction shall then be obtained as prescribed in 2.2.

4.2.13 If the mean pressure of the reference cartridges is not within ± 3500 PSI of the official assessed chamber pressure value, the test barrel shall be removed from the test and another test barrel substituted and another series of reference cartridges fired. If this firing fails to produce satisfactory pressure results, the cause thereof shall be identified and eliminated before resuming the test.

4.2.14 After the barrel has cooled to ambient temperature, warming (fouling) shots shall be fired in accordance with 4.2.1.

4.2.15 The test cartridges shall then be fired following the procedure prescribed in 4.2.2 through 4.2.11.

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4.2.16 The copper pressure cylinders used with the test cartridges shall then be measured and the measurements recorded in accordance with 3.1.2. The chamber pressure correction obtained with the reference cartridges shall be applied to the mean chamber pressure of the test cartridges, as prescribed in 2.2.

4.2.17 If any of the personnel conducting the test observes any abnormality tending to invalidate the velocity or pressure measurements, the circumstances shall be reported immediately through appropriate supervisory channels and the test suspended until instructions are received from proper authority.

4.3 Recording of Data.

4.3.1 Results of both, reference and test cartridges, should be recorded directly on the test sheet form. Velocities shall be recorded to the nearest FPS; pressures shall be recorded to the nearest 100 PSI.

4.3.1.1 Reference cartridges.- The individual velocities and chamber pressures, average velocity and chamber pressure, and extreme variation of velocity and chamber pressure.

4.3.1.2 Test cartridges.- The individual velocities and chamber pressures, average velocity and chamber pressure (not corrected), chamber pressure correction, average chamber pressure (corrected), extreme variation of the individual velocities and the individual chamber pressure and standard deviation of the individual chamber pressures.

4.3.1.3 Number and type of case casualties.

4.3.2 The following test weapon data shall be recorded on the test sheet:

a. Receiver number

b. Barrel number

c. Total number of cartridges fired in barrel.

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